5.6 DIFFERENCE BETWEEN CORRELATION AND REGRESSION

Correlation	Regression
1. It indicates only the nature and extent of linear relationship	It is the study about the impact of the independent variable on the dependent variable. It is used for predictions.
2. If the linear correlation is coefficient is positive / negative , then the two variables are positively / or negatively correlated	The regression coefficient is positive, then for every unit increase in x, the corresponding average increase in y is b_{YX}. Similarly, if the regression coefficient is negative , then for every unit increase in x, the corresponding average decrease in y is b_{YX}.
3. One of the variables can be taken as x and the other one can be taken as the variable y.	Care must be taken for the choice of independent variable and dependent variable. We can not assign arbitrarily x as independent variable and y as dependent variable.
4. It is symmetric in x and y, *ie.*, $r_{XY}=r_{YX}$	It is not symmetric in x and y, that is, b_{XY} and b_{YX} have different meaning and interpretations.

POINTS TO REMEMBER

- There are several types of regression - Simple linear correlation , multiple linear correlation and non-linear correlation.
- In simple linear regression there are two linear regression lines Y on X and X on Y.
- In the linear regression line $Y = a + bX + e$, where 'X' is independent variable, 'Y' is dependent variable, 'a' is intercept, 'b' is slope of the line and 'e' is error term.
- The point $(\overline{X},\overline{Y})$ passes through the regression lines.
- The " Method of least squares" gives the line of best fit.
- Both the regression lines have the same sign either positive of negative.
- The sign of the regression coefficient and the sign of the correlation coefficient is same.

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